HexManiacAdvance Mac Edition

Development accomplishments

Updated November 4, 2026

57	17,244	0
regression groups	assertions	failures / skipped groups

UI Improvements

- **Mac Edition welcome screen** - Added a responsive welcome screen with contributor credits, recognizable hyperlinks and a prominent Open ROM button.
- **Readable script display** - Aligned commands, dialogue, comments and highlighting with a shared monospaced font and corrected placeholder spacing.
- **Compact map tutorials** - Made tutorial cards readable on hover, reduced their footprint and kept map previews behind editor overlays.
- **Continuous encounter editing** - Kept focus and caret position stable while editing grass, surf, tree and fishing encounter tables.
- **Quick field replacement** - Made Tab and Shift Tab select complete short-field values without selecting entire script or dialogue editors.
- **Stable sprite selection** - Fixed large NPC sprite previews and recovered selector state after scrolling, history changes and reopening.
- **Fresh portrait and palette views** - Refreshed edited portrait choices and dismissed stale palette editors after Undo, reload or context changes.
- **Simpler toolbar** - Hid optional Claude and built-in emulator controls while retaining their implementations and external emulator access.

Mac Port Fixes

- **Control-click context menus** - Restored existing menus for tabs, warps, map arrows, hex content and palettes using Mac secondary-click conventions.
- **Finder integration** - Replaced Windows file-location launching with Finder reveal for the owning ROM file.
- **Remembered emulator choice** - Made Play launch a selected external emulator such as mGBA, remembering its application path and supporting Browse.
- **Focused clipboard commands** - Kept Command copy, cut and paste on the focused text or palette target instead of falling through to binary editing.
- **Safe field shortcuts** - Kept Delete, Backspace and arrow keys inside focused map-event fields instead of deleting or panning the event.
- **Control-click eyedroppers** - Made Control-click sample sprite colors and block tiles without entering the painting path.
- **Palette color entry** - Restored propagation of named swatch colors and hexadecimal color edits into palette data.
- **Map entry stability** - Removed the cold-map UI-thread error and focus-outline animation crash.
- **Restored editor controls** - Fixed rotated map arrows and restored specialized trainer, sprite and palette templates.
- **Dropped HMA scripts** - Restored drag-and-drop routing so HMA script files apply to the ROM instead of being handled as ordinary documents.

Map Editing Improvements

- **Trackpad map panning** - Added Option-drag panning that preserves the current brush and finishes correctly when Option is released first.
- **Placed-block selection** - Restored Mac Control-drag selection of placed blocks while preserving their collision and elevation data through saves and history.
- **Brush placement preview** - Added a translucent preview of the selected block pattern at its stamp destination.
- **Mixing bag instructions** - Corrected Mac help for Command-click bag membership, clearing the bag and activating random painting.
- **Permission-safe painting** - Preserved movement permissions during bag flood-fill and single-block rectangle painting.
- **New object scripts** - Restored the script-creation button so an object can receive an allocated starter script and attached pointer.
- **Stable NPC previews** - Eliminated complex hover-preview flicker and refreshed summaries after dialogue edits, Undo and Redo.
- **Interrupted gesture cleanup** - Stopped ghost painting and panning after capture loss, and kept interrupted strokes separate in Undo history.
- **Reliable same-cell edits** - Allowed painting, brush replacement and permission changes at the same cell after Undo or Redo.
- **Reveal mode cleanup** - Cleared transient Space reveal when keyboard focus, tabs or editing contexts change.
- **Trackpad-friendly zoom buttons** - Added centered minus and plus buttons that use the existing zoom levels and disable at their limits.
- **Map dimensions at a glance** - Displayed map size in tiles, native pixel dimensions on hover and clearly labeled cursor coordinates in the status bar.
- **Direct map PNG export** - Saved map images to MapImages beside the ROM with Overwrite, Save Updated Copy and Cancel choices for existing files.
- **NPC template sprite choice** - Added a bounded pre-placement sprite picker that remembers its independent choice and toolbar icon for the open ROM session.
- **Bounded tileset previews** - Kept blockset previews within the map viewport and limited standalone rendering to valid block counts.
- **Stable tileset selection** - Fixed the map-header dropdown crash while preserving immediate pointer writes, fresh previews and Undo/Redo.
- **Safe border access** - Rejected invalid border coordinates before calculating ROM addresses to protect neighboring data.
- **Consistent nine-grid permissions** - Applied explicit movement-permission choices to both inferred and fallback brush corners.

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Scripting Improvements

- **Automatic dialogue scaffolds** - Inserted an automatic text pointer and complete dialogue block with the caret ready for typing.
- **Automatic movement scaffolds** - Added complete movement blocks for player, camera, NPC and applymovement commands when the required operands are available.
- **Completion-aware scaffolding** - Ran dialogue and movement assistance after accepting a command suggestion as well as typing the command directly.
- **Flag and variable assistance** - Restored existing unused-ID helper buttons and Command U routing for eligible flag and variable commands.
- **Accurate autocomplete placement** - Corrected caret mapping and command replacement across multiline scripts while preserving explicit hexadecimal values.
- **Visible dialogue warnings** - Made overflow warnings and script errors visibly underline the affected text without adding popups or automatic wrapping.
- **Usable suggestion lists** - Fixed mouse acceptance and dismissed stale suggestions after outside clicks, focus changes, detachment or context replacement.
- **Aligned trainer warnings** - Corrected trainer move-token validation and warning spans while preserving existing matching and legality rules.

Under the Hood Improvements

- **Safer file saves** - Staged file replacements and preserved open unsaved documents when ROM or metadata saving fails.
- **Validated binary patches** - Validated patch structure and UPS checksums before changing ROM bytes, preserving existing edits after rejected IPS or UPS imports.
- **Validated project imports** - Rejected incompatible layouts, invalid values and lossy text before applying edits, with rollback if an import fails.
- **Recoverable project exports** - Published consistent table exports with owned-file manifests, writer coordination and recovery from interrupted publication.
- **Safer external reloads** - Handled replacement and rapid external saves more reliably while protecting local unsaved work.
- **Protected pending script source** - Bound deferred edits to their original context and accounted for pending source during Save, Save As, Close and reload.
- **Atomic script compilation** - Rolled back failed script and resource compilation across bytes, pointers, relocation, metadata and history while retaining drafts and diagnostics.
- **Mac diagnostics and updates** - Pointed crash/help links and compatible-release checks at the renamed Mac Edition repository while retaining upstream references where appropriate.

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Optimizations

- **Script highlighting** - Reduced the recorded 2,000-line highlighting workload from about 197 to 7.8 ms and temporary allocation from 21.6 to 3.3 MB per edit.
- **Script argument lookup** - Reduced a 2,000-command deferred update from 2,795 to 90 ms in the repeated-argument benchmark.
- **Script width layout** - Coalesced ten queued width calculations into one, reducing the measured 500-line burst from 39.6 to 3.3 ms.
- **NPC hover lookup** - Reduced empty-tile lookup on the 32-NPC fixture from about 4.51 to 0.065 ms by avoiding unrelated event-editor construction.
- **Map zoom rendering** - Removed all 70 unnecessary bitmap publications in the zoom-only benchmark while preserving rendered pixels.
- **Table navigation** - Reduced the measured binding and layout time for 100 complex navigation switches from 924 to 675 ms.
- **Portrait preparation** - Reduced 100 candidate-preparation operations from 77.4 to about 3.0 ms while preserving image freshness.
- **Trainer move validation** - Reduced validation of 100 six-member custom teams from 83.1 to 15.0 ms.
- **Item usage searches** - Reduced a measured delayed map-usage query from 225 to 148-160 ms by reusing script outcomes within that query.

Verification Improvements

- **Complete Mac regression runner** - Passed 57 groups and 17,244 assertions with zero failures and zero skipped groups in the last completed Mac run.
- **Real editor workflow coverage** - Expanded synthetic UI coverage for gestures, clipboard routing, suggestions, image previews, persistence and history.
- **Portable engine test hosts** - Enabled the original unit and integration test hosts on Mac with explicit optional-ROM skips and recorded legacy expectation differences.
- **NPC and tileset regressions** - Added 44 NPC-template assertions, 199 tileset-selection assertions and 160 bounded-preview assertions to the complete suite.

How to read the metrics

Performance figures are recorded component or workload measurements, not an overall app speedup or native interaction-latency guarantee; timings cannot be added, and temporary allocation is not retained RAM.

Evidence: CodexChanges entries 001-125 and the performance reports under tests; the latest Mac regression run is recorded at checkpoint 03c69932, while the separate upstream suites retain their own results and optional-fixture skips.

This sheet covers implemented work; the tutorial village, playable patch and future guides remain in development.